

Yoojin Lee 이유헌

☎ (+82) 10-2031-2992 | ✉ yjin1014@snu.ac.kr | 🌐 www.linkedin.com/in/yoojin-lee1014

Research Interests

- Translational immunoengineering, designing biomaterial based strategies to mimic and reprogram immune heterogeneity for cancer therapy.
- Biomaterial and microengineered platforms to modulate immune cell function through integrated chemical and mechanical cues.
- Tumor immunology and the tumor microenvironment, engineering immunosuppressive cells to restore effective antitumor immunity.
- Mechanical dynamics regulating immune cell decision making and phenotypic transitions.

Education

Seoul National University

M.S. In Interdisciplinary Program In Bioengineering

Seoul, South Korea

Aug. 2023 - Feb. 2026

Korea University

B.S. In School Of Biomedical Engineering

Seoul, South Korea

Mar. 2019 - Aug. 2023

Research Experience

Biomaterials For Immuno-Engineering Laboratory, Seoul National University

Seoul, South Korea

Researcher

Sep. 2023 - current

- Advisor Prof. Junsang Doh
- Characterized phenotype-dependent phagocytic behaviors of macrophages, gaining expertise in immune cell phenotyping, engineering platforms (nanogrooves for elongation, microwells for 1:1 interactions), and understanding how mechanical and chemical cues regulate activation.
- Developed a human cell based microphysiological system integrating immune cells for immunotherapy evaluation, focusing on system level immune regulation and cytokine responses.

Single-molecule Biophysics Lab, POSTECH

Pohang, South Korea

Research Intern (Advisor Prof. Jong-Bong Lee)

Jan. 2023 - Feb. 2023

- Engineered gene edits using CRISPR-Cas9 (knockout and knock-in) and validated outcomes through Nested PCR, T7E1 assay, resulting in hands-on expertise in genome editing workflows.
- Delivered LifeAct-fluorescent protein plasmids via lipofection and electroporation, enabling visualization of F-actin structures.
- Applied HILO microscopy to characterize cytoskeletal organization and analyzed single-cell fluorescent labeling patterns, contributing to insights on actin dynamics.

Laboratory Of Gastric Cancer Biology, Seoul National University

Seoul, South Korea

Research Intern (Advisors Prof. Han-Kwang Yang, Prof. Hyuk-Joon Lee, And Prof. Do Joong Park)

Jul. 2022 - Aug. 2022

- Cultured gastric cancer cell lines (SNU-1, NCI-N87) to test anticancer drug efficacy, resulting in recognition of cell line-specific responses and the need for tailored experimental approaches.
- Measured HER2 expression in gastric cancer cell lines using qPCR, Bradford assay, and Western blot, gaining integrated skills in gene-protein correlation analysis.
- Assisted in establishing patient-derived xenograft (PDX) models via subcutaneous implantation of GIST tissue into mice, building capacity for in vivo experimental techniques and understanding of translational cancer research models.

Biomedical Polymer Laboratory, POSTECH

Pohang, South Korea

Research Intern (Advisor Prof. Won Jong Kim)

Jan. 2022 - Feb. 2022

- Formulated lipid nanoparticles (LNPs) using microfluidic chip systems, gaining practical biomaterial handling skills and insight into formulation effects on immune cell interactions.
- Reviewed key literature on biomaterials and cancer immunotherapy, deepening knowledge of nanoparticle-based drug delivery and immune cell engineering for cancer treatment.

Conferences

Immunoreceptor Signaling And Therapeutic Applications, Cold Spring Harbor Asia

Awaji, Japan

Poster Presentation

Nov. 2025

- Presented a poster titled "Quantitative analysis of the phagocytic behaviors of macrophages in various biochemical/biophysical environments"

Korean Association Of Immunologists (KAI)

Seoul, South Korea

Poster Presentation

Apr. 2025

- Presented a poster titled "Characterization of phenotype-dependent diverse phagocytic behaviors of macrophages"

World Biomaterials Congress 2024

Conference Participation

- Attended sessions on biomaterials for immunotherapy

Daegu, South Korea

Jun. 2024

Extracurricular Activities

BIUS (Band Club, Department Of Biomedical Engineering, Korea Univ.)

Founder & Keyboardist

- Founded the department's first band club to rebuild student community after COVID-19, recruiting members across multiple year levels.
- Organized and performed in a live concert, coordinating rehearsals, set list planning, and event logistics.

Seoul, South Korea

Mar. 2022 - Mar. 2023

BEAT (Academic Club, Department Of Biomedical Engineering, Korea Univ.)

Core Member

- Presented literature reviews on various nanoparticles used in cancer treatment during weekly seminars
- Wrote an internal review report on current therapeutic strategies for gastric cancer
- Engaged in academic discussions and peer feedback sessions to deepen understanding of biomedical topics

Seoul, South Korea

Mar. 2022 - Dec. 2022

Student Council Member

Core Member

- Organized departmental volunteer events and academic gatherings for students
- Disseminated official departmental announcements and event information to students
- Assisted in coordinating communication between students and faculty

Seoul, South Korea

May. 2019 - Feb. 2020

Patents

2025

Co-inventor, Method for Quantitative Analysis of Macrophage Phagocytic Activity Using Multi-fluorescent Labeling and Flow Cytometry (patent application, Korea Intellectual Property Office, pending)

Licenses & Certifications

BIML 2025 Workshop, Korean Society For Bioinformatics

Participant

- Participated in the offline BIML 2025 workshop featuring Dr. Jong-Eun Park (KAIST), focused on best practices in single-cell data analysis

Seoul, South Korea

May. 2025

60Th Public Lecture, Institute For Data Innovation, Seoul National University

Participant

- Completed training course on Python-based data processing, visualization, regression analysis, and basic machine learning

Seoul, South Korea

Jan. 2025

Workshop On Techniques In Laboratory Animals

Bio Medical Center For Animal Resource Development, SNU

Seoul, South Korea

Jul 2022

Skills

Laboratory Skills

Molecular biology techniques (PCR, T7E1 assay, gel electrophoresis)

Genome editing and transfection (CRISPR-Cas9 knockout/knock-in, electroporation, lipofection)

Protein analysis (Western blot, ELISA)

Cell biology and imaging (immunofluorescence, live-cell and confocal imaging, flow cytometry)

Microphysiological systems and biomaterials (MPS chip handling, hydrogel loading and gel patterning)

Nanomaterials and microscopy (nanogroove and microwell fabrication, SEM, TEM)

Programming

Python, MATLAB, R

Languages

Korean: Native, English: Proficient (TOEIC 955, TOEFL 98, IELTS 7.0)

Honors

DOMESTIC

Fall 2023 **Academic Excellence Scholarship** Seoul National University (10% of semester fees)

Seoul, South Korea